Assessment 2 — Marking Rubric

DSB102 Introduction to Machine Learning — S2 2026

Weighting Summary

The project is worth **60 marks (60% of the total unit grade)**.

Component	Marks	% of unit grade	Assessed by
Group report (Tasks 1–3)	40	40%	Lecturer + teaching team
Presentation (incl. Q&A)	10	10%	Tutor + one observer
Individual reflection	10	10%	Lecturer + teaching team
Total	**60**	**60%**	

The report and presentation are group marks; every member normally receives the same mark for them. See the Project Brief, “Group Contribution”, for the limited circumstances in which an individual’s mark may be adjusted.

The Two Project Themes

Students are assigned to groups by the teaching team; **within your group, you choose which one of the two themes to work on**, and confirm your theme and member roles **by the end of Week 9**. Both themes carry the same total marks and the same rubric — choose based on interest, not perceived difficulty.

Theme	Client	Problem
Classification — Customer Churn	Sydney Bank	Predict which customers are at risk of churn
Regression — Sales Forecasting	Retail Dynamics	Forecast weekly sales to support inventory planning

Full briefs: Sydney Bank Customer Churn and Retail Dynamics Weekly Sales — provided separately, one per theme.

How This Is Marked

This project is assessed as if the group were a consulting team presenting findings to a real client. The rubric rewards **quality of thinking, honesty of evaluation, and clarity of communication** — not which algorithm was chosen.

A well-justified simple model outscores a complex model presented without explanation. An honest error analysis outscores an inflated one. A clear recommendation outscores a technically impressive report with no actionable conclusion.

Use only the modelling methods taught in DSB102. Classification theme (Customer Churn): logistic regression, k-nearest neighbours, decision trees. Regression theme (Sales Forecasting): linear regression, decision trees, random forests. Methods from outside the unit — including time-series methods such as

ARIMA, exponential smoothing or moving averages for the Sales theme — are not part of this assessment and should not be used. A submission that uses one is marked on the in-unit analysis and reasoning it still contains, with no credit for the out-of-unit method.

Report length: maximum 12 pages, in total — including the front page, all figures, and references. No code appendix: the analysis is accessed via the report's Google Colab link, not pasted into the document. Content beyond page 12 is not marked. Full format requirements (front page contents, Colab link): see the Project Brief, "Report Format".

Component 1: Group Report (40 marks)

Each task's exact sub-parts and mark split are specified in the project brief for your theme. Task 1 and Task 3 are identical across both themes; Task 2 differs only in how the final 2 marks are allocated (see Task 2 below).

Task 1 — Understand the Landscape (8 marks)

Patterns across ≥3 conditions with numbers (2) · ≥2 quantified hypotheses (2) · assumptions listed (2) · follow-up questions named (2). Exact wording: see your theme's brief. Spotting and standardising the dataset's formatting inconsistency is not a separately-marked item — it supports the assumptions here and the Task 2 data-preparation step.

What we look for: specific numbers for every claim (not "sales vary by promotion" but "sales average X units with a display vs Y without"); the formatting inconsistency noticed and a fix stated; assumptions that are concrete and numbered; sensible, specific follow-up questions.

Task 2 — Plan Your Analytical Approach (10 marks)

Target/problem type stated (1) · features justified (1.5) · data preparation approach (1.5) · baseline + ≥2 models named with reasons (2) · train/test split + metric justified (2) · final 2 marks (theme-specific): the **Classification theme** splits these as risks (1) + responsible use of protected attributes (1); the **Regression theme** puts all 2 on the business constraint / cost asymmetry, with risks expected in the plan but not separately scored. See your theme's brief.

What we look for: the target and problem type stated explicitly and correctly (regression vs classification); features justified individually, not just listed; a named baseline and ≥2 named candidate models with reasons; train/test split and metric(s) justified in business terms; the specific business constraint addressed concretely.

Task 3 — Execute the Analysis (22 marks)

What Did We Find? (8) · What Should [the client] Do? (8) · What Could the Impact Be? (6).

What we look for: ≥2 models compared against the baseline with actual numbers on the chosen metric(s); the preferred model justified on more than "lowest error" — fit for the business problem; **three** specific, actionable Factor → Effect → Action chains, one for each of the top-three factors the report names, that the client could start next month; a concrete impact estimate with a stated assumption; at least one honest, specific limitation.

Report task total: 8 + 10 + 22 = 40.

Report Presentation

Beyond the task content itself, we also expect: a short executive summary (≤200 words) written for a non-technical decision-maker; clear structure that moves through the three tasks; figures that are numbered, captioned, and referenced in the text; and professional writing that makes good use of the 12-page limit.

There's no separate mark bucket for this — the report total is exactly Task 1 + Task 2 + Task 3 = 40 — but presentation quality affects how each task's work is judged.

Code and Google Colab link (required). The analysis behind the report must be uploaded as a Google Colab notebook, and the report must include a link to it. The link must be shared so the whole marking team can open it (sharing set to "anyone with the link can view"), and the notebook must **run end to end** on the provided dataset without errors.

If the Colab link is missing, inaccessible, or does not run: apply a **flat 5-mark deduction** to the report total, **once** (not per task or per claim). Then mark each task normally, treating any number or result that cannot be reproduced from working code as unsupported — it is marked on the written reasoning alone, which is where the "unsupported numbers score low" effect comes from. There is no further deduction beyond the flat 5 marks and the task marks themselves. The report mark cannot fall below 0.

Component 2: Presentation + Q&A (10 marks)

Delivered live, **during your Week 13 workshop — recorded video is not accepted**. All group members present some part of the talk — aim for roughly equal speaking time where possible, though this isn't a strict requirement. Every member must be prepared to answer Q&A questions regardless of which part they presented. **All group members must attend the entire Week 13 workshop session**, not just arrive for their own group's presentation slot.

Format: 10 minutes presentation + 3 minutes Q&A per group (13 minutes total). Groups that run over on the presentation will be stopped; undelivered content cannot be assessed. Slides are submitted the night before — see the Project Brief, "Key Deadlines".

Content & narrative (4 marks): covers approach, results, and recommendation, matching the report's Task 1/2/3 structure — every slide earns its place, not a notebook walkthrough.

Delivery & timing (3 marks): confident delivery from knowledge, not reading; stays within time; slides clear, legible, uncluttered.

Q&A (3 marks): all members can explain the reasoning behind choices made in the report — including members who didn't present that section.

Presentation total: 4 + 3 + 3 = 10.

Full presentation requirements (required slides, preparation guidance): see the Project Brief, "Presentation Requirements" section.

Component 3: Individual Reflection (10 marks — assessed individually)

Each student submits their own reflection (up to 300 words) **individually via Canvas** — due **Friday 23 October 2026** (with an automatic 48-hour extension to Sunday 25 October 2026), see the Project Brief, "Key Deadlines" — this is not the group leader's responsibility and is not part of the group's report submission.

Contribution clarity (4 marks): a specific personal contribution described with concrete examples (what was built, written, or decided), including an honest account of a difficulty faced and how it was handled.

Team process insight (3 marks): genuine insight into how the team worked together — a specific decision, disagreement, or division of labour — not just "we communicated well."

Learning (3 marks): covers all three: (a) a specific, non-generic insight from the Industry Forum that could not have been written without attending; (b) what was learned about ML or the project itself; (c) if GenAI was used, one concrete experience — what was prompted, what was produced, and what was verified or changed. If GenAI was not used, state this and describe how you independently checked your work.

Reflection total: 4 + 3 + 3 = 10.